MSG No. 112.265 $P_C\bar{4}2c$ [Type IV, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `222.`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,3,4]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[5,6,7,8]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[9,10,11,12]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[13,14,15,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `222.`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,2,3,4]`
2	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[5,6,7,8]`
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[9,10,11,12]`
4	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[13,14,15,16]`

Table 3: Wyckoff site: `4c`, site symmetry: `-4..`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,4,5,6]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[2,3,7,8]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[9,12,13,14]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[10,11,15,16]`

Table 4: Wyckoff site: `4d`, site symmetry: `-4'..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,4,13,14]`
2	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,3,15,16]`
3	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[5,6,9,12]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[7,8,10,11]`

Table 5: Wyckoff site: `8e`, site symmetry: `.2.`

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{4}\right]$	`[1,3]`
2	$\left[0,\ -y,\ \frac{1}{4}\right]$	`[2,4]`
3	$\left[y,\ 0,\ \frac{3}{4}\right]$	`[5,8]`
4	$\left[-y,\ 0,\ \frac{3}{4}\right]$	`[6,7]`
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[9,11]`
6	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	`[10,12]`
7	$\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[13,16]`

8	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{3}{4}\right]$	[14,15]

Table 6: Wyckoff site: 8f, site symmetry: .2.

No.	position	mapping
1	$\left[0,\ y,\ \frac{3}{4}\right]$	[1,3]
2	$\left[0,\ -y,\ \frac{3}{4}\right]$	[2,4]
3	$\left[y,\ 0,\ \frac{1}{4}\right]$	[5,8]
4	$\left[-y,\ 0,\ \frac{1}{4}\right]$	[6,7]
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{3}{4}\right]$	[9,11]
6	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	[10,12]
7	$\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	[13,16]
8	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{1}{4}\right]$	[14,15]

Table 7: Wyckoff site: 8g, site symmetry: 2..

No.	position	mapping
1	$[0,\ 0,\ z]$	[1,4]
2	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	[2,3]
3	$[0,\ 0,\ -z]$	[5,6]
4	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	[7,8]
5	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	[9,12]
6	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[10,11]
7	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	[13,14]
8	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[15,16]

Table 8: Wyckoff site: 8h, site symmetry: 2..

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	[1,4]
2	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[2,3]
3	$\left[\frac{1}{2},\ 0,\ -z\right]$	[5,6]
4	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	[7,8]
5	$\left[\frac{1}{2},\ 0,\ z\right]$	[9,12]
6	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	[10,11]
7	$\left[0,\ \frac{1}{2},\ -z\right]$	[13,14]
8	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[15,16]

Table 9: Wyckoff site: 8i, site symmetry: 2'..

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	[1,12]
2	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	[2,11]
3	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[3,10]

continued ...

Table 9

No.	position	mapping
4	$\left[\frac{1}{4}, \frac{3}{4}, z\right]$	[4,9]
5	$\left[\frac{1}{4}, \frac{1}{4}, -z\right]$	[5,14]
6	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	[6,13]
7	$\left[\frac{3}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	[7,16]
8	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	[8,15]

Table 10: Wyckoff site: 16j, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$\left[x, -y, \frac{1}{2}-z\right]$	[2]
3	$\left[-x, y, \frac{1}{2}-z\right]$	[3]
4	$[-x, -y, z]$	[4]
5	$[y, -x, -z]$	[5]
6	$[-y, x, -z]$	[6]
7	$\left[-y, -x, z+\frac{1}{2}\right]$	[7]
8	$\left[y, x, z+\frac{1}{2}\right]$	[8]
9	$\left[x+\frac{1}{2}, y+\frac{1}{2}, z\right]$	[9]
10	$\left[x+\frac{1}{2}, \frac{1}{2}-y, \frac{1}{2}-z\right]$	[10]
11	$\left[\frac{1}{2}-x, y+\frac{1}{2}, \frac{1}{2}-z\right]$	[11]
12	$\left[\frac{1}{2}-x, \frac{1}{2}-y, z\right]$	[12]
13	$\left[y+\frac{1}{2}, \frac{1}{2}-x, -z\right]$	[13]
14	$\left[\frac{1}{2}-y, x+\frac{1}{2}, -z\right]$	[14]
15	$\left[\frac{1}{2}-y, \frac{1}{2}-x, z+\frac{1}{2}\right]$	[15]
16	$\left[y+\frac{1}{2}, x+\frac{1}{2}, z+\frac{1}{2}\right]$	[16]